

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Assessment Tool 1 – Application based questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
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Section / Batch:	B.Tech IT	Date:	30/07/2026

Instructions to Students

- Identify the relevant computer architecture concepts, functional units, or performance parameters required to solve the given application-based problem.
- Analyse the interaction between CPU, memory, bus structures, and I/O components in the given scenario.
- Apply appropriate instruction execution cycles, addressing modes, and architectural concepts to determine the correct solution.
- Compute performance metrics such as CPI, clock cycles, CPU execution time, speedup, or benchmarking measures using appropriate formulas and calculations.
- Interpret the calculated results and explain their impact on overall computer system performance.

QUESTIONS

1. A user reports that a computer becomes slower when several applications are opened simultaneously. To analyse the system behaviour, observe the CPU and Memory usage before and after launching multiple applications. Analyse the observed resource utilization to determine how the CPU, memory, and I/O devices interact during program execution and identify the functional unit that is most likely responsible for the performance change.
2. A user wants to compare the performance of two computers to determine which system is expected to execute programs more efficiently. To compare the systems, observe the CPU model, Base Speed, Number of Cores, and Number of Logical Processors for each system. Analyse the observed processor specifications to determine which computer is expected to provide better performance, and justify your conclusion by relating the specifications to CPU execution time and overall processor performance.
3. A programmer wants to understand how an 8085 microprocessor executes instructions during program execution. To examine the instruction execution process, write and execute a program using an 8085 simulator. Observe the changes in the Program Counter (PC), Accumulator (A), and Flag Register after each instruction. Analyse how the fetch, decode, and execute stages contribute to producing the final result.

Code of Conduct

I Abdul Wahid / 192321079 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
ProblemSolving Approach	20	Logical, wellstructured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, wellorganized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Q. No	Understanding of Problem Context (20)	Application of Concepts (30)	ProblemSolving Approach (20)	Accuracy of Solution (20)	Clarity (10)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 30	/ 20	/ 20	/ 10	/ 100

Course Coordinator

Faculty Incharge

① Performance when multiple applications are opened.

Observation:

Before opening multiple applications, CPU and memory usage are lower. After opening several applications, CPU and memory usage increase. Then open multiple applications and observe the CPU, memory and I/O usage again.

Analysis:

Each application needs CPU time to execute instructions and memory to store its data and instructions. When many applications run simultaneously, the processor has to handle more tasks and memory usage increases. If CPU utilization becomes very high, the system takes more time to process the tasks.

Conclusion:

The CPU / Processor is the functional unit most likely responsible for the performance change when its utilization becomes high.

② Comparing the performance of two computers

Observation:

Check and record the following specifications for both computers:

- * CPU model.
- * Base speed.
- * Number of Cores.
- * Number of Logical Processors.

Then compare the values of both systems.

Analysis:

A higher CPU base speed generally allows instructions to be processed faster. More cores allow the processor to handle more tasks simultaneously, while more logical processors can improve multitasking and parallel execution.

Conclusion:

The computer having the better combination of CPU speed, number of cores and logical processors is expected to execute programs more efficiently. It should generally provide better overall processor performance and lower execution time.

③ 8085 ~~Block~~ Instruction Execution.

Observation:

Write the 8085 program in a simulator and execute it one instruction at a time. After every instruction, record the values of the program counter (PC), Accumulator (A) and Flag register.

Analysis:

The 8085 first fetches the instruction from memory. It then decodes the instruction to understand what operation must be performed and finally executes it. During execution, the PC moves to the next instruction, the Accumulator changes according to the operation, and the flags change when the result affects them.

Conclusion:

By observing the PC, Accumulator and Flag Register & after each instruction, we can understand how the 8085 executes the program step-by-step and how the final result is produced.